Order of Operations (BIDMAS)

Chapter 1: Numbers — Homework Solution Key

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Fundamental Rule: BIDMAS / BODMAS Priority Order

To evaluate mathematical expressions correctly, operations must be carried out in the following strict hierarchy:

1. **B**rackets / Parentheses: Evaluate expressions inside brackets ( ) first.
2. **I**ndices / Powers / Roots: Evaluate exponents (x^2), square roots ($\sqrt{x}$), etc.
3. **D**ivision & **M**ultiplication: Perform left-to-right in the order they appear.
4. **A**ddition & **S**ubtraction: Perform left-to-right in the order they appear.

QUESTION 1 (April/2018/Paper 1/ Q.4) [1 Mark]

A teacher asks her class to work out the answer to:

$$8 + 12 \div 4$$

Mike says that the answer is 5. He is wrong.
Explain why Mike is wrong.

Model Answer & Marking Breakdown

Step-by-Step Solution:

1. **Identify Mike's error:** Mike calculated from left to right without applying BIDMAS:
$$(8 + 12) \div 4 = 20 \div 4 = 5 \quad \textbf{(Incorrect priority)}$$
2. **Apply correct order of operations (Division before Addition):**
   - First perform division: $12 \div 4 = 3$.
   - Then perform addition: $8 + 3 = 11$.

Correct Statement / Explanation for Mark:

> **"Mike added $8 + 12$ before dividing by 4. According to BIDMAS, division must be done before addition ($12 \div 4 = 3$, so the correct answer is $8 + 3 = 11$)."**

Mark Scheme:

[1 Mark] Correct explanation explicitly stating that division takes priority over addition (or giving the correct answer 11 with $12 \div 4 = 3$ shown first).

QUESTION 2 (Oct./2019/Paper 1/ Q.3) [1 Mark]

Work out:

$$5 \times (4^2 + 2 - 12)$$

Model Answer & Marking Breakdown

Step-by-Step Solution:

1. **Evaluate inside the Brackets first:**
   - Inside $(4^2 + 2 - 12)$, evaluate the Index/Power first: $4^2 = 16$.
   - The bracketed expression becomes: $(16 + 2 - 12)$.
2. **Complete addition and subtraction inside brackets (left to right):**
   - Addition: $16 + 2 = 18$.
   - Subtraction: $18 - 12 = 6$.
3. **Multiply by the factor outside the bracket:**

$$5 \times 6 = \mathbf{30}$$

Mark Scheme:

[1 Mark] Final correct value of **30** (with working shown).

QUESTION 3 (April/2020/Paper 1/ Q.1) [1 Mark]

Work out the value of:

$$5^2 + \sqrt{121}$$

Model Answer & Marking Breakdown

Step-by-Step Solution:

1. **Evaluate the Index/Power term:**

$$5^2 = 5 \times 5 = 25$$

2. **Evaluate the Square Root term:**

$$\sqrt{121} = 11 \quad (\text{since } 11 \times 11 = 121)$$

3. **Add the two terms together:**

$$25 + 11 = \mathbf{36}$$

Mark Scheme:

[1 Mark] Final correct answer **36**.

QUESTION 4 (Oct./2022/Paper 1/ Q.26) [2 Marks]

Work out:

$$(23 - 3 \times 3)^2$$

Model Answer & Marking Breakdown

Step-by-Step Solution:

1. **Evaluate inside the Brackets first:**
   - Multiplication takes priority over subtraction inside brackets:

$$3 \times 3 = 9$$

   - The bracket simplifies to:

$$23 - 9 = 14$$

2. **Apply the outer Index/Power (Square the result):**

$$14^2 = 14 \times 14 = \mathbf{196}$$

Mark Scheme:

[M1] Method mark for correctly evaluating inside the bracket to get 14 (e.g. showing $23 - 9 = 14$ or 14^2).

[A1] Accuracy mark for final correct value of **196**.

QUESTION 5 (April/2019/Paper 1/ Q.20) [2 Marks]

Put **one set of brackets** in each calculation to make the answer correct.

(a) $4 + 9 \times 6 - 4 = 22$

(b) $24 \div 12 - 8 + 2 = 4$

Model Answer & Marking Breakdown

Step-by-Step Solution & Testing Placement:

(a) Target answer $= 22$:

- Test placing brackets around $(6 - 4)$:

$$4 + 9 \times (6 - 4) = 4 + 9 \times 2 = 4 + 18 = \mathbf{22} \quad \checkmark$$

Answer (a): $4 + 9 \times (6 - 4) = 22$

(b) Target answer $= 4$:

- Test placing brackets around $(12 - 8 + 2)$:

$$24 \div (12 - 8 + 2) = 24 \div (4 + 2) = 24 \div 6 = \mathbf{4} \quad \checkmark$$

Answer (b): $24 \div (12 - 8 + 2) = 4$

Mark Scheme:

[1 Mark] For part (a) correctly bracketed: $4 + 9 \times (6 - 4) = 22$.

[1 Mark] For part (b) correctly bracketed: $24 \div (12 - 8 + 2) = 4$.

QUESTION 6 (April/2016/Paper 1/ Q.19) [2 Marks]

Put **one pair of brackets** in each statement to make it correct.

(a) $10 - 4 \div 2 + 18 = 21$

(b) $7 \times 3 + 1 + 2 = 30$

Model Answer & Marking Breakdown

Step-by-Step Solution & Testing Placement:

(a) Target answer $= 21$:

- Test placing brackets around $(10 - 4)$:

$$(10 - 4) \div 2 + 18 = 6 \div 2 + 18 = 3 + 18 = \mathbf{21} \quad \checkmark$$

Answer (a): $(10 - 4) \div 2 + 18 = 21$

(b) Target answer $= 30$:

- Test placing brackets around $(3 + 1)$:

$$7 \times (3 + 1) + 2 = 7 \times 4 + 2 = 28 + 2 = \mathbf{30} \quad \checkmark$$

Answer (b): $7 \times (3 + 1) + 2 = 30$

Mark Scheme:

[1 Mark] For part (a) correctly bracketed: $(10 - 4) \div 2 + 18 = 21$.

[1 Mark] For part (b) correctly bracketed: $7 \times (3 + 1) + 2 = 30$.

QUESTION 7 (Core) [1 Mark]

Calculate the value of:

$$2 + 8(40 - 5)$$

Model Answer & Marking Breakdown

Step-by-Step Solution:

1. **Evaluate inside the Brackets first:**

$$40 - 5 = 35$$

The expression becomes: $2 + 8(35)$ or $2 + 8 \times 35$.

2. **Perform Multiplication before Addition:**

$$8 \times 35 = 280$$

3. **Perform Addition:**

$$2 + 280 = \mathbf{282}$$

Mark Scheme:

[1 Mark] Final correct answer **282**.

— End of Homework Model Answer Key —